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1 Introduction

TP53 is a well-established tumour suppressor gene and one of the most frequently mutated genes in human cancers. APR246 is an experimental therapeutic agent that reactivates p53, thereby inhibiting cancer cell growth and inducing apoptosis. This analysis aims to explore the transcriptional effects of APR246 in the p53-mutated breast cancer line BT459. The dataset consists of paired-end RNA-seq data generated from BT459 cells treated with APR246 or DMSO (the control), with three biological replicated per condition (1). Several Linux-based workflows were used to process the sequencing data. Raw reads were assessed for quality using FastQC (sbatch 01), followed by adapter trimming with BBMap (sbatch 02) and a second quality check on trimmed files (sbatch 03). An index was generated (sbatch 04), the reads were aligned to the reference genome (sbatch 05), and gene counts were generated (sbatch 06) and annotated (sbatch 07). Additional commands such as sacct -j ¡jobID¿ were used to monitor job status, and more slurm-¡jobID¿.out was used to view batch job output.

2 Interpretation of FastQC Files

2.1 Per Base Sequence Quality

A summary of the FastQC reports for the untrimmed and trimmed sequences is seen in Table 1. The per base sequence quality module evaluates the quality score distribution at each base position across all reads. The FastQC uses a box-whisker plot to determine whether base qualities are sufficiently high, with scores categorised as good (green), moderate (orange) or poor (red). Warnings occur is the lower quartile for any base is less than Q10 or the median quality for any base is less than Q25, while failures occur if the lower quartile for any base is less than Q5, or if the median in less than Q20. All the sequences in this analysis showed a good per sequence base quality, indicating extremely low error rates and allowing for reliable alignment and downstream quantification (2).

2.2 Per Sequence GC Content

The per GC content module checks whether the %GC distribution of the reads resembles a normal distribution. Deviations from the expected shape may indicate a technical bias or contamination. Warnings occur if greater than 15% of the reads deviate from the expected distribution, and failures occur is greater than 30% of reads deviate. A high %GC content often indicates potential adapter dimers. Most of the sequences in this dataset showed a moderate GC content deviation, suggesting mild but unproblematic technical deviation (2)

2.3 Sequence Duplication Levels

The sequence duplication levels module measures the proportion of reads are that exact duplicates. High duplication indicates low library complexity and may result in technical issues such as PCR over-amplification or biological factors such as a highly abundant RNA transcripts, which is common in RNA-seq analysis. Warnings occur is greater than 20% of reads are duplicates, and failures occur is greater than 50% of reads are duplicates. All sequences received a bad duplication report, suggesting low library complexity likely due to PCR amplification, highly abundant transcripts, or the presence of repeated sequence types (3).

Table 1: Summary of FASTQC outcomes across raw and trimmed BT459 RNA-seq files

Dataset group	Per base quality	GC content	Duplication levels
APR246 raw	All good	Mostly moderate; one good	All bad
DMSO raw	All good	Mostly moderate; two good	All bad
APR246 trimmed	All good	Mostly moderate; one good	All bad
DMSO trimmed	All good	Mixed moderate/good	All bad

3 Interpretation of Summary Plots

3.1 Plot 1: Total number of mapped and unmapped reads per sample

Read mapping is the process of aligning millions of short read sequences from the RNA to a reference genome or transcript. All the samples contain approximately twenty-three to thirty million reads, which is a typical depth for bulk RNA sequencing (4). The APR246 biological replicates (n=3) show slightly higher read counts (27-30 million reads) compared to the DMSO replicates (n=3) (23-28 million reads), as seen in Figure 1. The unmapped portion is extremely small for all samples, indicating high mapping efficiency and good read quality.

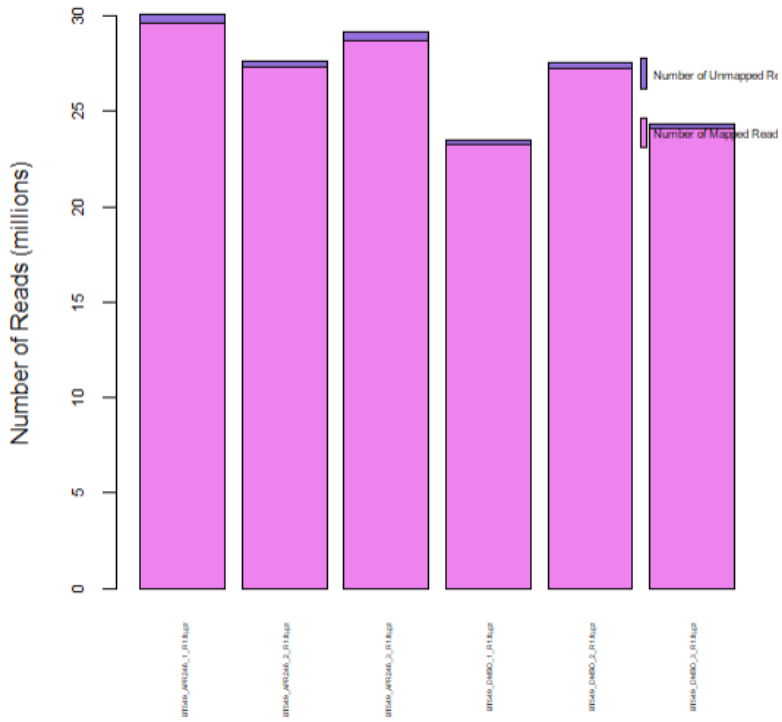

Figure 1: Total number of mapped and unmapped reads per sample. All samples show high sequencing depth ($\sim 22 - 30$ million reads) with very low unmapped fractions, indicating efficient alignment and good overall read quality.

3.2 Plot 2: Library size of each sample

Library sizes represent the total number of sequences reads, reflecting the overall sequencing depth for each sample (5). The library sizes range from approximately seventeen million to twenty-three million reads as seen in Figure 2, which falls within the range for a typical RNAseq library size of greater than twenty million aligned reads (6).

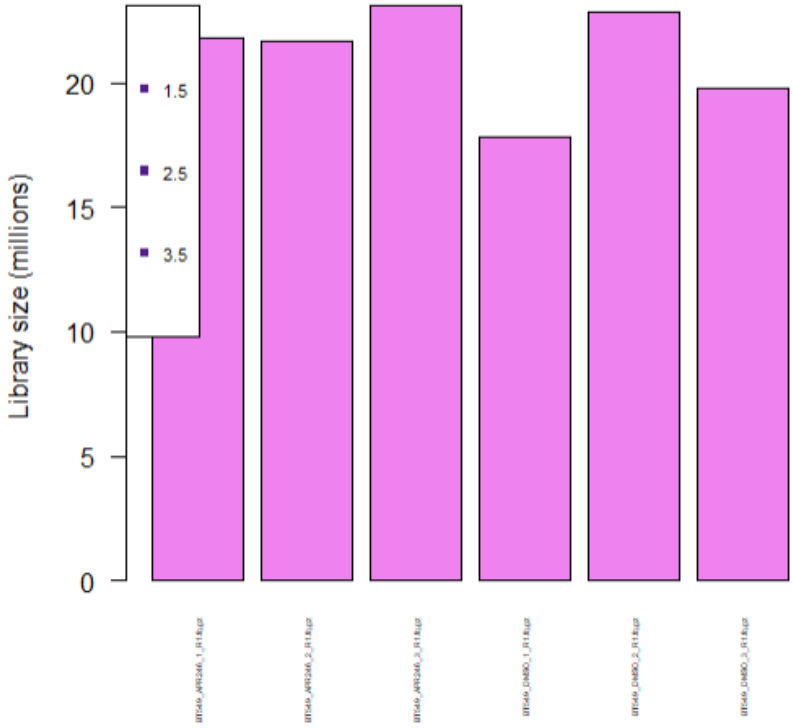

Figure 2: Library size of each sample. Library sizes range from Ĩ7-23 million counts and don't convey any distinct outliers, indicating comparable sequencing depth across replicates for APR246 (n=3) and DMSO (n=3).

3.3 Plot 3: MDS Plot showing sample distances based on gene expression

The MDS plot shown in Figure 3, has a clear separation between APR246 and DMSO samples along Principal Component 1 (65%), deeming it as a major source of variation. APR246 samples tend to cluster tightly together on the left side of the plot, indicating high similarities between replicates (n=3). DMSO replicates (n=3) cluster on the right side apart from one replicated (DMSO_2) which is slightly separated along Principal Component 2 (25%). None of the samples appear as an outlier and it's clear that clustering is driven by treatment effect rather than technical effects.

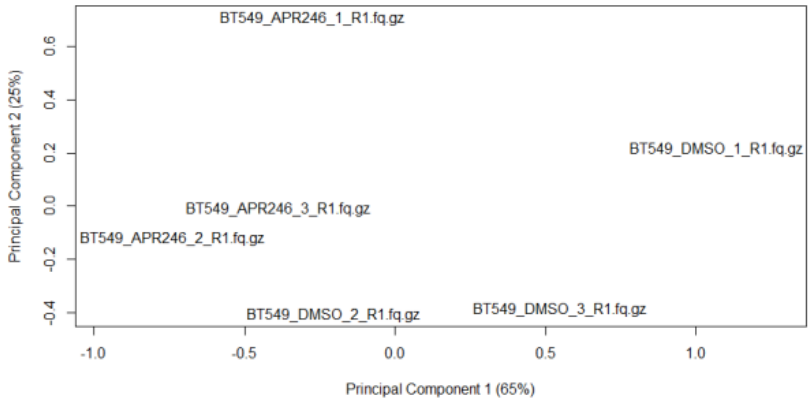

Figure 3: MDS Plot showing sample distances based on gene expression. APR246 and DMSO samples form two distinct clusters along principal component 1, indicating a treatment-driven variation

4 Association of Differentially Expressed Genes with Oncogenes

4.1 Obtaining Differentially Expressed Genes

To identify differentially expressed genes between APR246-treated and DMSO control samples, the experimental groups were defined in a design matrix, allowing the model to recognise which samples belonged to each condition. A contrast matrix was then generated to define the comparison of interest (APR246-DMSO), meaning that the positive log fold change values represent genes upregulated in APR246, while negative values indicate downregulation or upregulation in DMSO. The data was subsequently normalised using the limma-voom pipeline, which adjusts for library size differences, ensuring that expression measurements are comparable across samples for accurate statistical testing

After fitting the linear model and obtaining basic statistics, the results were filtered by adjusted p-value to retain only statistically significant genes. Within this set, the top five upregulated and top five downregulated genes were selected based on the largest absolute log fold changes. A volcano plot as seen in Figure 4 was generated to visualise these results.

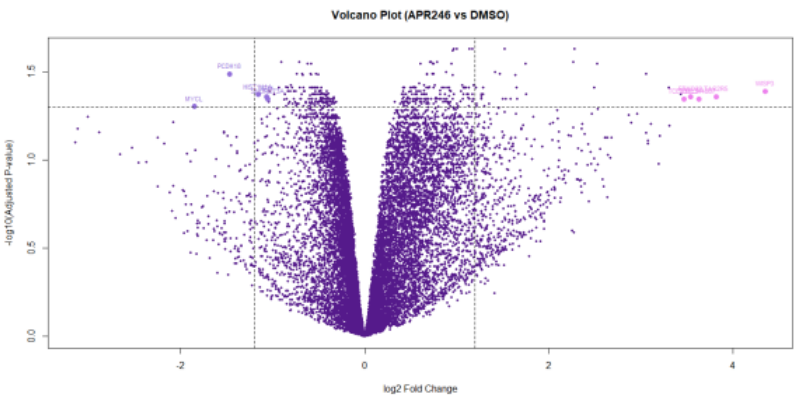

Figure 4: Volcano plot showing differentially expressed genes between APR46 and DMSO. Significantly upregulated genes in APR246 (right side, coloured lilac) include WISP3, TAS2R5, LOC44887, FBXO39, and TCEB3CL2. Significantly downregulated genes in APR246 (left side, coloured periwinkle) include MYCL, PCDH18, HIS1H1A, TET1, and SNORD73A.

4.2 Top 5 Upregulated Genes

Table 2 shows the top five most upregulated genes in APR246-treated cells compared to DMSO, all with positive log fold changes between 3.4 and 4.3 and statistically significant adjusted p-values (p ¡ 0.05). These genes include WISP3, TAS2R5, LOC44887, FBXO39, and TCEB3CL2.

To assess whether any of the top five upregulated gene were known oncogenes, each gene was cross-referenced with OncoKB, a clinically curated precision oncology knowledge database developed by Memorial Sloan Kettering Cancer Centre. OncoKB provided evidence-based annotations on cancer-driven genes, their biological effects, and their clinical relevance. Genes listed in this database are considered to have well-supported role in tumour development (7). Among the top upregulated genes, only WISP3 was identified in OncoKB and classified likely oncogenic, prompting further investigation.

WISP3 has a complex role in cancer. It can function as either an oncogene or a tumoursuppressor depending on the tumour type. In certain cancers, including subsets of breast and ovarian cancer, WISP3 is overexpressed and promotes proliferation, suggesting oncogenic activity. In contrast, in inflammatory breast cancer and some colorectal cancers, WISP3 is downregulated and exerts tumour-suppressive effects (8). The remaining upregulated genes – FBXO39, LOC344887, and TAS2R5 – were not listed as oncogenes in OncoKB, but there is some evidence about their emerging oncogenic roles.

FBXO39 has been reported to promote proliferation, migration, and survival in several cancers, including osteosarcoma, glioma, and cervical squamous cell carcinoma, indicating potential oncogenic activity even though it's not yet recognised as a canonical oncogene (9). LOC344887, despite being a pseudogene – and therefore unable to function as a classical protein-coding oncogene (10)– has been shown to exert cancer-promoting regulatory effects in hepatocellular carcinoma and gallbladder cancer through pathways such as STAT3/HMGA2 (11). Conversely,

TAS2R5 is not considered oncogenic as it encodes a bitter taste receptor, and activation of TAS2R5 family members is typically associated with tumour-suppressive effects, including reduced proliferation and increased apoptosis (12).

In conclusion, among the top five upregulated genes in APR246-treated BT459 cells, only one gene (WISP3) is recognised in OncoKB as a potential oncogene. The remaining upregulated genes are not established oncogenes, and some such as TAS2R5 are associated with tumoursuppressive or anti-proliferative functions. This overall expression pattern suggests that APR246 does not drive oncogenic transcriptional responses in this analysis, which is consistent with the drug exerting its intended anti-tumour effects rather than promoting tumorigenesis.

Table 2: Top 5 Upregulated Genes

GeneID	Chr	Length	Symbol	Type	logFC	AveExpr	t	P.Value	adj.P.Val
6836	6	1408	WISP3	protein-coding	4.3510	-3.1115	5.8472	0.000253	0.040951
54429	7	1150	TAS2R5	protein-coding	3.8176	-2.8457	5.3826	0.000457	0.043833
344887	3	11312	LOC44887	pseudo	3.6325	-2.9426	5.2901	0.000516	0.045166
162517	17	1667	FBXO39	protein-coding	3.5422	-3.5149	5.3840	0.000457	0.043833
100506888	18	1877	TCEB3CL2	protein-coding	3.4701	-3.5493	5.2760	0.000521	0.045166

4.3 Top 5 Downregulated Genes

Table 3 shows the top five most downregulated genes in APR246-treated cells compared to DMSO, all with negative log fold changes between (-)1.0 and 1.8 and statistically significant adjusted p-values (p ¡ 0.05). These genes include MYCL, PCDH18, HIS1H1A, TET1, and SNORD73A.

Each of the top 5 downregulated genes was cross-referenced with OncoKB to determine whether any had recognised oncogenic roles. MYCL and TET1 were the only genes identified in the database. MYCL belongs to the MYC proto-oncogene family, one of the most wellestablished groups of cancer-driving transcription factors (13). TET1, however, exerts both oncogenic and tumour-suppressive functions depending on the disease context. It operates as a tumour suppressor in many solid cancers – such as breast, gastric, colorectal, and liver tumours – evidence shows that specific isoforms or hypoxic conditions can shift TET1 toward oncogenic behaviour (14).

The remaining downregulated genes are not recognised oncogenes. PCDH18 is widely considered a tumour suppressor, frequently silence in colorectal and other cancers, where its loss promotes tumour progression (15). HIST1H1A is not a classical oncogene, but its dysregulation can enhance tumour aggressiveness, particularly in prostate cancer, suggesting a tumour-promoting rather than oncogenic role (16). Finally, SNORD73A is a small nuclear RNA and therefore can't function as a protein-coding oncogene, though dysregulated snoRNAs can control tumorigenesis by modulating oncogenic pathways (17). Overall, among the top downregulated genes,

MYCL is the only wellestablished oncogene, with the others functioning primarily as tumour suppressors.

In conclusion, among the top five downregulated genes in APR246-treated BT459 cells, two genes (MYCL and TET1) were identified in OncoKB. MYCL is a well-established oncogene belonging to the MYC proto-oncgene family, and its strong downregulation is consistent with reduced tumour-promoting activity. TET1, although listed in the OncoKB database primarily functions as tumour suppressor in many solid cancers and only exhibits oncogenic properties under specific conditions such as hypoxia and isoform-specific shifts. The remaining downregulated genes are not recognised oncogenes and are associated with tumour-suppressive or regulatory roles. This overall expression pattern indicates that APR246 downregulates a key oncogene and drives tumour-suppressive transcriptional responses, which is consistent with the drug exerting anti-tumour effects rather than promoting tumorigenesis.

Table 3: Bottom 5 Differentially Expressed Genes

GeneID	Chr	Length	Symbol	Type	logFC	AveExpr	t	P.Value	adj.P.Val
4610	1	495	MYCL	protein-coding	-1.8529	-0.6159	-4.7760	0.001339	0.049767
54510	4	5906	PCDH18	protein-coding	-1.4707	-4.6953	-7.2715	0.000495	0.032687
3024	6	781	HIST1H1A	protein-coding	-1.1602	2.7298	-5.5283	0.000379	0.042606
80312	10	9584	TET1	protein-coding	-1.0709	2.8809	-5.3710	0.000645	0.043941
8944	4	65	SNORD73A	snoRNA	-1.0551	1.3496	-5.0722	0.000689	0.045181

5 Influence of Enriched Pathways in Breast Cancer

5.1 Generating Pathway Enrichment Analysis

In this dataset, the top enriched pathways – cell cycle, microRNAs and FoxO signalling – are all involved in the development and progression of breast cancer, as seen in Figure 5. The adjusted p-values and counts of the enriched pathways are shown in Table 4.

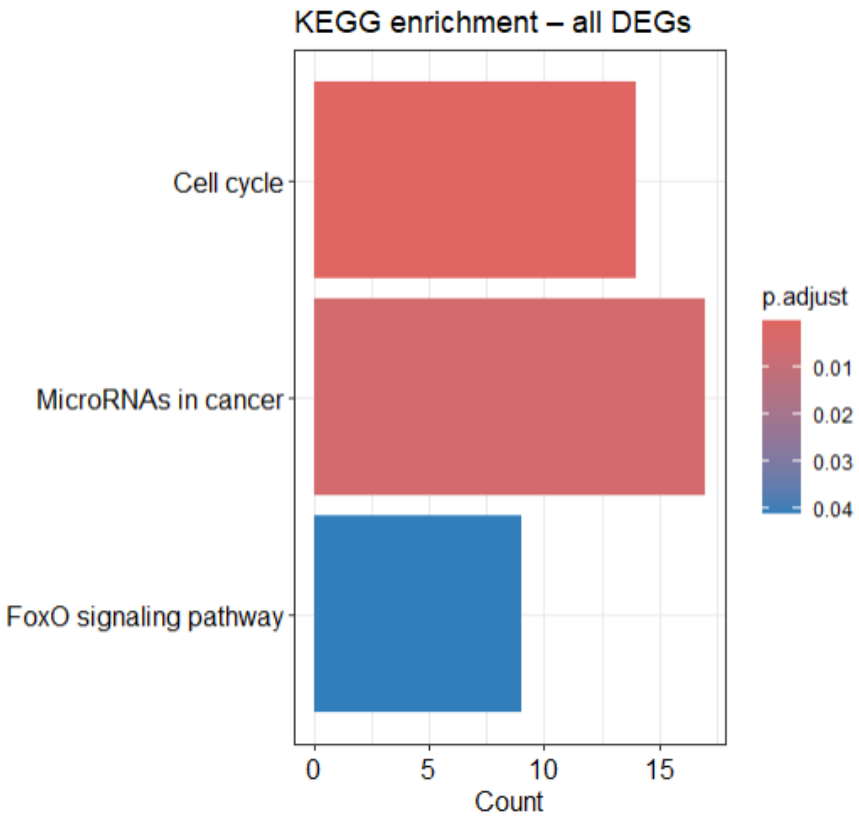

Figure 5: KEGG pathway enrichment analysis of differentially expressed genes. The most significantly enriched pathways include the cell cycle regulation, microRNAs in cancer, and the FoxO signalling pathways.

Table 4: Enriched KEGG Pathways

Category	ID	Description	p.adjust	Count
Cellular Processes	hsa04110	Cell cycle	0.000142	14
Human Diseases	hsa05206	MicroRNAs in cancer	0.004837	17
Environmental Info Processing	hsa04068	FoxO signalling pathway	0.041087	9

5.2 The Influence of the Cell Cycle on Breast Cancer

In normal breast tissue, the cell cycle is a process that controls the growth, DNA replication and division of cells through a series of checkpoints. These checkpoints make sure that damaged DNA within the cell is repaired before it continues to next stage, otherwise the damaged cell is pushed towards apoptosis instead of division (18). Progression through G1, S2, G2 and M phases is driven by fluctuating levels of cyclins and their cyclin-dependent kinases (CDKs). For example, Cyclin D1 and CDK4/6 promote passage through the G1/S checkpoint, while tumour suppressors like p53 and RB1 function as brakes, halting the cycle in response to stress or DNA damage to maintain genomic stability (19). Even though the oestrogen receptor (ER) and PI3K

signalling normally promote growth signals, in a normal setting they are still constrained by these core safety mechanisms (20).

In breast cancer, this system is rewritten to favour proliferation. Amplification and overexpression of cyclin such as CCND1 and CCNE1, combined with overactivation of CDK4/6, push cells through the G1/S checkpoint even when they shouldn't divide (19). CCND1 amplification is particularly common in hormone receptor-positive tumours, linking ER signalling directly to abnormal cell cycle progression, while CCNE1 amplification is more typical of triple-negative and HER2-positve breast cancers (18). Simultaneously, loss-offunction mutations or deletions in p53, RB1 and PTEN remove key inhibitory signals, allowing cells with DNA damage to continue through the cell cycle and accumulate further mutations (18). Hyperactive PI3K signalling and other oncogenic pathways further support survival growth, so that the normal balance between proliferation, repair and cell death is lost, and uncontrolled division becomes a prominent feature of breast (20).

5.3 Influence of microRNAs on Breast Cancer

MicroRNAs (miRNAs) add an extra post-transcriptional layer of control to this system. Under normal conditions, miRNAs are small non-coding RNAs that calibrate the expression of large sets of genes by binding to target mRNAs and repressing their translation or promoting their degradation (21). This allows them to regulate pathways involved in cell proliferation, differentiation, apoptosis, and response to stress in a very precise way.

In cancer miRNA expression become dysregulated, and this contributes directly to tumour behaviour. Some miRNAs act as oncogenic miRNAs (oncomiRs) and are overexpressed in tumours when they suppress tumour-suppressor genes and promote proliferation, invasion and therapy resistance (22). Others normally act as tumour suppressors as their loss remove brakes on oncogenes and pro-survival pathways (21). Due to the fact that a single miRNA can target many genes, these changes can reshape entire networks, including those controlling the cell cycle and apoptosis (22). At the same time, the stability of miRNAs means that they can be picked up in serum or plasma, and multiple circulating miRNA signatures have been proposed for early detection, prognosis and prediction of recurrence in breast cancer patients (23). This makes miRNAs both active regulators of disease and useful biomarkers for tracking it.

5.4 Influence of FoxO Signalling on Breast Cancer

FoxO signalling sits at the intersection between these pathways. In normal cells, FOXO transcription factors (such as FOXO3a) help maintain homeostasis by promoting cell cycle arrest, apoptosis and resistance to oxidative stress (24). When active in the nucleus, FOXO3a can upregulate CDK inhibitors like p21 and p27, downregulate Cyclin D1, and induce proapoptotic genes, effectively reinforcing checkpoints that prevent damaged cells form dividing (25). Its ac-

tivity is tightly regulated by upstream pathways such as PI3K-Akt, which phosphorylates FOXO proteins and drives them out of the nucleus, and by stress-activated pathways that can reactivate them under conditions like high ROS (24).

In breast cancer this balance becomes more complex. In one respect, loss of activation of FOXO3a removes an important tumour-suppressive influence, allowing ER-driven and PI3K-driven proliferation to proceed unchecked. This forces FOXO3a activation in ER-positive breast cancer cells which has been shown to suppress proliferation and block oestrogen dependent tumour growth by upregulating CDK inhibitors, reducing Cyclin D1 and ERregulated survival genes (25). Conversely, in certain contexts FOXO1 and FOXO3 can promote invasion, metastasis or drug resistance, for example by inducing matrix metalloproteinases or supporting survival under therapeutic stress (24). FOXO expression and activity are also influenced by specific miRNAs, linking FoxO signalling to the broader miRNA regulatory network (24). Overall, in breast cancer, the FoxO pathway can either help re-establish cell cycle control and apoptosis or, if overtaken by oncogenic signalling, contribute to adaption and progression of the cancer (26).

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